

Music Style Transfer and Deep Generative Genre Remastering

A 50-minute technical lecture followed by a DGGR case study

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Part I general lecture: problem framing, theory, representations, models, and evaluation

Part II DGGR case study: what we built, what worked, what still fails, and embedded audio

🔗 Goal

Treat this as a lecture first. We will start from the field itself, then use our repo as a concrete implementation example near the end.

🎧 Media note

Audio clips are embedded and also attached to the PDF. Acrobat-compatible viewers handle inline playback most reliably.

Talk Structure

Part I: general lecture

Motivation, definitions, historical context, core model families, evaluation criteria, and the main failure modes in music style transfer.

Part II: DGGR case study

Architecture, lab-by-lab progress, saved-run evidence, audio demos, creative elements, and realistic next steps.

Guiding question

How do we change genre while keeping the identity of the song recognizable?

Answering that requires a representation story, a conditioning story, a synthesis story, and a coherence story.

Our project can then be evaluated as one specific answer to that larger systems question.

Part I
General Lecture

Lecture section

Motivation: Why This Problem Is Hard

Audio is unforgiving

Small errors in phase, transients, or continuity are immediately audible. Artifacts that might be tolerated in images are distracting in music.

Music is multilevel

Melody, rhythm, harmony, instrumentation, arrangement, and production are entangled in the same signal.

The central mistake in weak systems

A shallow timbre swap often changes surface texture without changing the deeper musical organization. The result sounds processed, not re-conceived.

What a stronger system must do

1. Isolate content from style.
2. Encode a target genre in a controllable form.
3. Synthesize realistic audio.
4. Preserve coherence over long horizons.

Lecture framing grounded in Dai et al. (2018), Hung et al. (2019), and Yang et al. (2019).

Problem Statement and Success Criteria

$$\begin{aligned} x_{\text{src}} &\longrightarrow \hat{x}_{\text{tgt}} \\ \text{subject to } &\text{content}(\hat{x}_{\text{tgt}}) \approx \text{content}(x_{\text{src}}) \\ &\hat{x}_{\text{tgt}} \in \mathcal{M}_{\text{genre}=\text{tgt}} \end{aligned}$$

Operational goal

The output should sound as if the source had originally been arranged or performed in the target genre, not merely post-processed into it.

Three simultaneous objectives

- Content preservation
- Style/genre fidelity
- Acoustic realism

Long-form adds a fourth

Coherence across chunk boundaries, phrase structure, and minute-scale texture stability.

Representation Levels in Music ML

Waveform level

Highest fidelity and direct physical meaning, but editing semantics in raw 1D audio is difficult.



Time-frequency level

Spectrograms, log-mel features, chroma, onsets, and beat grids make structure and texture more visible to the model.



Symbolic or structured level

MIDI, chord sequences, note rolls, and phrase segments are good for long-range organization but weak for sonic realism.

Design lesson

Strong systems usually combine levels. Structure is often easier to preserve in higher-level representations, while realism is recovered with a learned decoder, vocoder, or codec.

Why this matters later

Our repo follows this pattern directly: log-mel features for analysis, codec latents or mel diffusion for generation, and external decoders for waveform fidelity.

What Do We Mean by Content, Style, and Genre?

Content

Melody, harmonic movement, rhythmic backbone, phrase placement, and whatever listeners still identify after instrumentation changes.

Style

Timbre, articulation, groove feel, density, attack/decay behavior, production texture, and arrangement cues.

Genre

A broader cultural and statistical manifold. No single acoustic feature defines it, but enough structure is learnable for classification and conditioning.

$$x \mapsto (z_c, z_s), \quad \text{genre} \approx \mathcal{G}(z_s, \text{context}), \quad \hat{x} = D(z_c, z_{s, \text{target}})$$

Why Genre Is Especially Difficult

Ill-defined but still learnable

Genre depends on cultural convention, geography, era, production context, and listener expectation. That makes it semantically fuzzy, but not unusable in practice.

What machine learning exploits

Shared rhythmic, spectral, timbral, and arrangement statistics allow genre classifiers to work well even though genre itself is not reducible to one intrinsic feature.

Research consequence

A generator should not treat genre as a one-hot texture label. It needs a richer target signal that captures a region of stylistic behavior.

Evaluation consequence

If the output is realistic but listeners would not place it in the target genre, the transfer still failed.

This slide follows Aucouturier and Pachet (2003) plus later genre-recognition work such as Ghosal and Kolekar (2018).

How We Evaluate a Style-Transfer System

Axis	Question	Typical measurement
Content preservation	Does the melody / structural identity survive?	latent similarity, pitch correlation, MPS-style metrics
Style fidelity	Does the output move toward the target style?	classifier confidence, probe accuracy, centroid alignment
Audio realism	Does it sound plausible as audio?	adversarial critics, MR-STFT, FAD-like proxies, listening
Long-form coherence	Does it stay consistent over time?	boundary metrics, drift checks, human judgment

Important principle

No single metric is sufficient. The best numeric checkpoint is often not the best listening checkpoint, especially in generative audio.

Historical Evolution of the Field

Era	Typical approach	Main weakness
Early transfer	Direct signal mapping, hand-crafted transforms, image-style analogies on spectrograms	Weak semantic control and strong artifact risk
Disentanglement wave	Separate pitch, rhythm, timbre, or content/style in latent space	Leakage is hard to eliminate and harder to verify
High-fidelity generation	GANs, neural codecs, universal vocoders, diffusion models	Better realism, but training and control are more complex
Long-form systems	Hierarchical latents, source anchoring, overlap constraints, chunk policies	Minute-scale drift and accumulation remain difficult

Trend

The field moved away from single monolithic mappings toward staged systems with separate analysis, control, and synthesis modules.

Why that matters

Once the problem is seen as multistage, engineering decisions become explicit rather than hidden inside one giant black box.

Disentanglement as the First Core Tool

$$E(x) \rightarrow (z_c, z_s)$$

$$\mathcal{L}_{\text{content}} = \|z_c(x^{(a)}) - z_c(x^{(b)})\|_2^2$$

$$\mathcal{L}_{\text{style}} = \text{CE}(C_s(z_s), y_{\text{style}})$$

$$\mathcal{L}_{\text{adv}} = \text{CE}(C_c(\text{GRL}(z_c)), y_{\text{style}})$$

Interpretation

The content code is rewarded for staying stable across views and punished when style can be recovered from it. The style code is explicitly trained to remain style-bearing.

Role of GRL

The gradient reversal layer flips the adversarial signal so the encoder learns to hide style evidence from the content branch.

Examples of Disentanglement Architectures

DuoED / dual-encoder designs

One encoder specializes in pitch or content, another in timbre or style, and multiple decoders force each latent to carry interpretable information.

U-Net / skip-connection designs

Let structural information bypass the bottleneck while style is learned in the latent path. This helps preserve pitch but can limit edit strength.

Sequence models and analogy systems

GRU or VAE-based systems split latent dimensions into rhythm and pitch blocks so one can be changed while the other remains fixed.

Practical lesson

Architecture alone does not guarantee separation. You still need audits for style leakage, collapse, and downstream usefulness.

Target Style Spaces and Genre Blueprints

$$v_{\text{target}} = \text{normalize}([\alpha z_s \parallel \beta d])$$

$$\mu_g = \frac{1}{|S_g|} \sum_{x \in S_g} v(x)$$

Idea

The target genre is not a single example. It is a region or centroid in a style space built from many examples and optionally mixed with robust descriptors.

What makes a style space useful

Linear separability, stable centroids, good nearest-neighbor behavior, and resistance to outliers or dataset imbalance.

Failure mode

If the style space is weak, the generator either ignores it or learns only a shallow edit.

Conditioning Mechanisms: How the Target Signal Enters the Model

$$\text{FiLM}(F; c) = \gamma(c) \odot F + \beta(c)$$

$$\text{AdaIN}(F; s) = (1 + \gamma(s)) \odot \text{IN}(F) + \beta(s)$$

Concept

Rather than append the condition only at the input, modulation layers influence intermediate features throughout the network.

When FiLM is attractive

Use it when a condition should steer a feature hierarchy across many depths and timescales.

When separate style modulation helps

If melody/structure and timbre/texture should not compete for the same path, FiLM and AdaIN can be separated by role.

Generator Families: Strengths and Tradeoffs

Family	Strength	Main tradeoff
Waveform GANs	Fast parallel generation and strong local realism when stable	Phase and local coherence are difficult; training can be brittle
Spectral GANs	Better phase handling in structured frequency domains	Need a reliable inversion / vocoding path
Neural codec editors	Decoder acts as a realism prior; easier than raw waveform synthesis	The latent space can constrain edit magnitude if the translator is too conservative
Diffusion models	Stable optimization and rich controllability	Slower inference; long-form consistency is still hard

Field-level takeaway

There is no universally best generator. The choice depends on what matters more in the application: edit strength, fidelity, speed, or stability.

Diffusion Models and Guided Editing

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I)$$

$$v_t = \sqrt{\bar{\alpha}_t} \epsilon - \sqrt{1 - \bar{\alpha}_t} x_0$$

$$\hat{\epsilon}_{\text{cfg}} = \epsilon_{\theta}(x_t, t, \emptyset) + w \left[\epsilon_{\theta}(x_t, t, c) - \epsilon_{\theta}(x_t, t, \emptyset) \right]$$

Meaning of the equations

The forward process adds noise, the reverse model learns to remove it, and classifier-free guidance controls how hard the condition pulls the sample toward the target.

Why this is attractive for style transfer

Guidance gives a direct knob over edit strength, which is especially useful when content preservation and genre shift compete.

Long-Form Coherence: Why Generation Gets Harder Over Time

$$x_t = \alpha(t)x_0 + \sigma(t)z, \quad z \sim \mathcal{N}(0, I)$$

SDEdit intuition

Start from a noised version of the source rather than from pure noise. This preserves macro-structure while still allowing local restyling.

Chunking intuition

Long songs are often generated in overlapping windows. The hard part is keeping those windows semantically aligned rather than merely crossfaded.

Failure modes

Boundary seams, phrase drift, timbral inconsistency, warble accumulation, and eventual departure from the source melody.

Useful controls

Overlap locking, re-anchoring, source blending, hierarchical structure priors, and post-hoc seam diagnostics.

General Design Rules Before We Look at Our Project

Rule 1

Do not ask one network to discover content, style, and waveform realism from scratch unless the data and compute budget justify it.

Rule 2

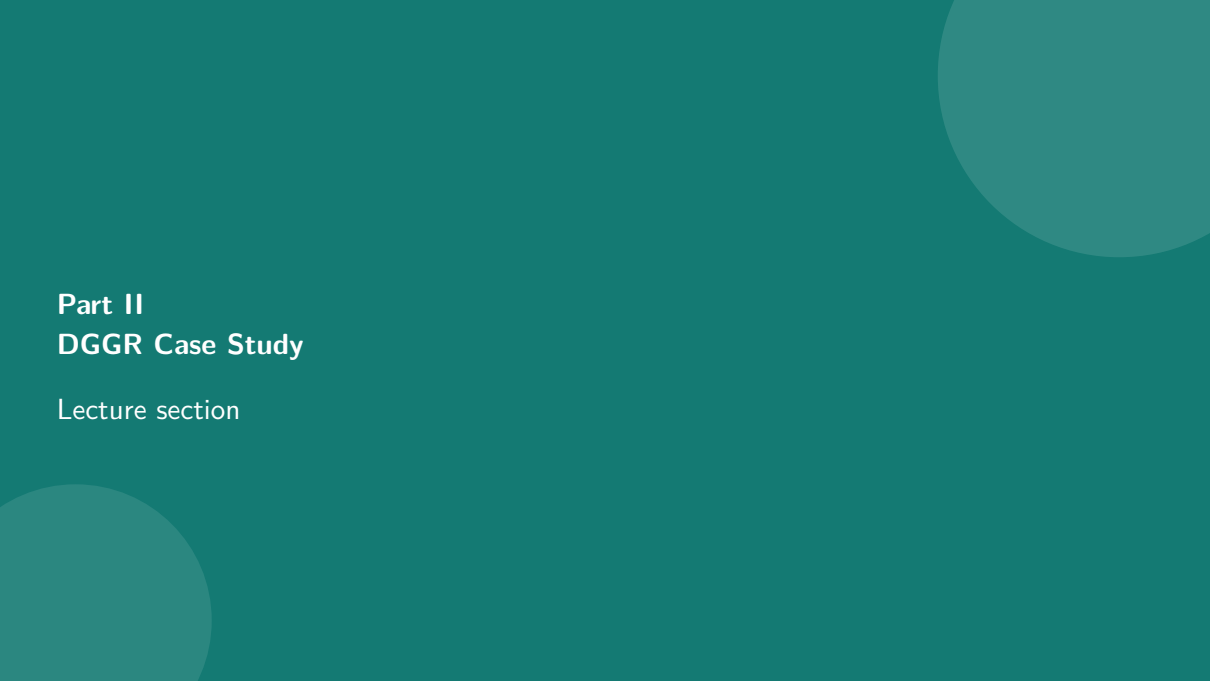
Measure each stage separately. A weak style space can make a strong generator look bad, and vice versa.

Rule 3

Listening must be in the loop. For music generation, the best scalar metric is often not the best sounding checkpoint.

Rule 4

Long-form success is a systems problem involving generation policy, overlap handling, stabilization, and evaluation, not just one model choice.



Part II

DGGR Case Study

Lecture section

DGGR: Project Goal and Current Status

Project thesis

Deep Generative Genre Remastering treats genre transfer as a staged remastering problem: deconstruct the source, define a target blueprint, reconstruct realistic audio, then extend that to long-form coherence.

Current repo-backed status

Labs 1 through 4 are implemented and have saved artifacts. Lab 5 is the planned human evaluation layer and baseline comparison study.

How this maps to the lecture

Lab 1 = disentanglement

Lab 2 = target style space

Lab 3 = reconstruction

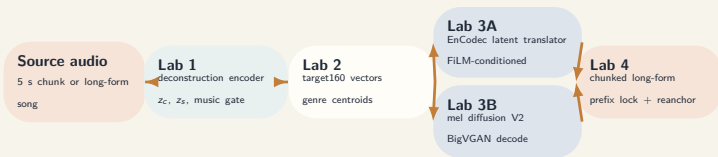
Lab 4 = long-form coherence

Lab 5 = human validation

What we will show

Concrete architectures, metrics from the repo, and audio clips generated from saved runs.

Full DGGR Pipeline



Why staged?

Each lab solves one of the general lecture problems explicitly instead of hoping one model learns all of them implicitly.

Best current branch

The codec translator is currently the strongest short-form result path. The diffusion branch is more ambitious and drives the long-form experiments.

Data Universe and Label Risks

What appears in the current workflow

Baroque/classical renders, hip-hop material, lo-fi material, open-domain CC0 music, and speech negatives for the music gate.

Observed genre buckets in the current Lab 2 run

baroque_classical: 411
cc0_other: 1200
hiphop_xtc: 1200
lofi_hh_lfbb: 1200

Critical caveat

Genre labels can accidentally encode dataset source. A model might learn recording or collection fingerprints instead of generalizable musical style.

Why we say this explicitly

Strong engineering presentations should surface confounds, not hide them. The repo docs already identify this as a real risk.

Lab 1: Deconstruction Encoder Architecture

Input

96-bin log-mel spectrogram from a fixed 5-second window



Backbone

3-block Conv2D encoder with stride-2 downsampling, global average pooling, and a shared projection layer



Heads

$z_c \in \mathbb{R}^{128}$, $z_s \in \mathbb{R}^{128}$, style classifier, adversarial style-from-content classifier, and a music gate head

Training curriculum

Phase 1 emphasizes content separation.

Phase 2 increases adversarial pressure.

Phase 3 sharpens the music gate and optionally uses a teacher anchor.

Why this lab matters

Every later module depends on the assumption that content and style have been separated well enough to recombine cleanly.

Lab 1: Objective, Thresholds, and Outcome

$$\mathcal{L}_{\text{Lab1}} = \lambda_c \mathcal{L}_{\text{content}} + \lambda_s \mathcal{L}_{\text{style}} + \lambda_a \mathcal{L}_{\text{adv}} + \lambda_g \mathcal{L}_{\text{gate}} + \lambda_t \mathcal{L}_{\text{anchor}}$$

Project thresholds

style accuracy ≥ 0.85

content leakage ≤ 0.15

gate ROC AUC ≥ 0.90

Best observed audit

style probe accuracy: **0.9417**

content leakage above baseline: **0.1083**

gate ROC AUC: **0.9299**

Interpretation

The lab clears its scientific gates. The remaining caution is calibration: an aggressive music-gate threshold can still trade recall for precision.

Lab 2: Building the Target160 Space

$$\text{target160} = \text{normalize}([2.0 \cdot z_s \parallel 1.0 \cdot d_{32}])$$

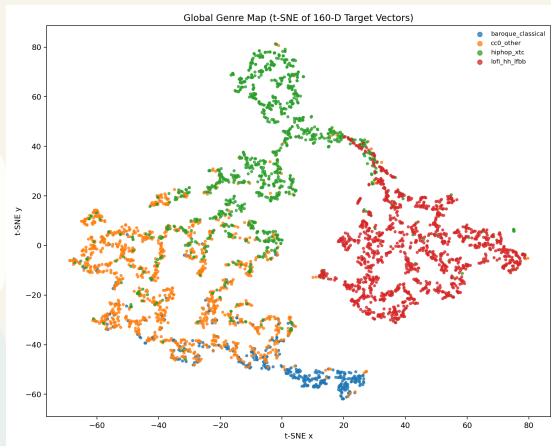
$$d_{32} = [\mu_{1:16}, \sigma_{1:16}]$$

Construction logic

The vector combines learned style information with simple mel-descriptor statistics, then builds per-genre centroids after inlier filtering.

Why not just use a class label?

A continuous target vector carries more structure than a one-hot tag and can support stronger conditional generation.



The visualization comes from current repo calibration artifacts, not from an external paper figure.

Lab 2: Validation and Interpretation

Current validation metrics

linear probe accuracy: **0.8554**

nearest centroid accuracy: **0.8514**

silhouette (cosine): **0.4939**

Completion gate

The lab threshold was silhouette ≥ 0.45 . The current run clears it with margin.

Why this matters downstream

A generator can only follow a style target if that target has stable geometry. Lab 2 is what makes the later conditioning story credible.

Presentation lesson

This is the best point in the talk to emphasize that representation quality is not separate from generation quality; it is upstream of it.

Lab 3A: Codec-Latent Translator Architecture

$$q_{\text{src}} \in \mathbb{R}^{128 \times T}, \quad c = [z_c \parallel z_{s,\text{tgt}} \parallel \eta], \quad \eta \in \mathbb{R}^{32}$$

$$q_{\text{hat}} = \begin{cases} q_{\text{src}} + s \tanh(r) & \text{residual mode} \\ r & \text{direct-output mode} \end{cases}$$

Implemented network

Conv1D in-projection, hidden width 256, 10 residual FiLM blocks with dilations [1, 2, 4, 8, 1, 2, 4, 8, 1, 2], GroupNorm, SiLU, then a Conv1D back to 128 codec channels.

Why EnCodec matters

The pretrained decoder acts as a realism prior. We edit a latent representation that already decodes to plausible audio instead of learning waveform physics from scratch.

Important design decision

Direct-output mode removes the identity shortcut that was capping style-shift magnitude in earlier residual runs.

Lab 3A: Loss Stack and Best Run

$$\mathcal{L}_{\text{codec}} = \lambda_{\text{adv}} \mathcal{L}_{\text{hinge-GAN}} + \lambda_{\text{fm}} \mathcal{L}_{\text{feat}} + \lambda_1 \|q_{\text{hat}} - q^*\|_1 \\ + \lambda_{\text{mrstft}} \mathcal{L}_{\text{MR-STFT}} + \lambda_c \mathcal{L}_{\text{content}} + \lambda_s \mathcal{L}_{\text{style}} + \lambda_p \mathcal{L}_{\text{push}}$$

Best saved run: run1055

MPS: **0.9565**

style confidence: **0.8940**

style accuracy: **0.9492**

Why this branch is strong

1. MERT probe embeddings improved target style conditioning.
2. Direct-output translation enabled stronger edits.
3. Explicit content loss kept the stronger edits from destroying melody.

Current conclusion

For short-form transfer, the codec branch is the strongest evidence path in the repo today.

Lab 3B: Diffusion V2 Architecture

Input tensor

$[B, 15, 80, 432] = 1 \text{ noisy mel} + 12 \text{ chroma} + 1 \text{ onset} + 1 \text{ beat-grid channel}$

Output tensor

$[B, 1, 80, 432]$ velocity/noise prediction for DDIM sampling

UNet backbone

channel progression $[64, 128, 256, 256]$, two residual blocks per level, self-attention at low resolutions, EMA shadow model for sampling stability

$$\text{cond}_{tc} = f_t(t) + f_c(z_c)$$

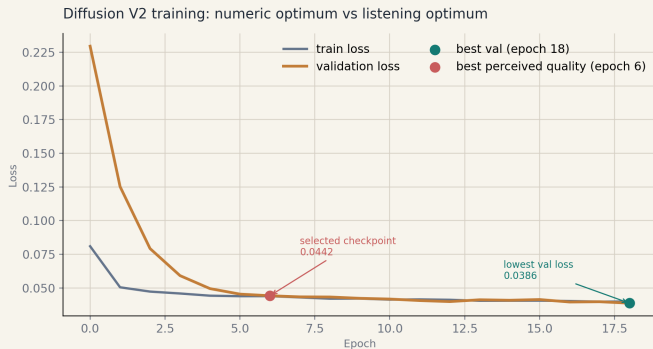
$$s = f_s(z_s)$$

$$h' = \text{AdaIN}(\text{FiLMResBlock}(h, \text{cond}_{tc}), s)$$

Design rationale

Time and content share a FiLM path for structure; style gets its own AdaIN path so timbre control does not overwhelm melody control.

Diffusion Training Behavior and Checkpoint Selection



Observed behavior

Validation loss kept improving through epoch 18, but the team selected epoch 6 as the best-sounding checkpoint for actual generation.

Interpretation

Later checkpoints appear numerically better but perceptually more averaged. This is why listening checkpoints belong in the workflow, not just in the final demo.

Lab 4: Long-Form Coherence Method

$$x_t^{(k)} = \alpha(t)x_0^{(k)} + \sigma(t)\epsilon$$
$$\text{prefix}\left(x_t^{(k)}\right) \leftarrow \text{tail}\left(x_t^{(k-1)}\right)$$

Implemented recipe

1. Split the song into overlapping chunks.
2. Forward-diffuse each chunk from the source mel instead of pure noise.
3. At every DDIM step, lock the new chunk's overlap prefix to the previous chunk's overlap tail.
4. Use re-anchoring and source blending to reduce drift.

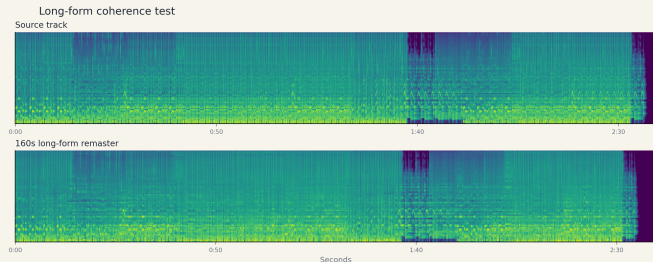
Why not just crossfade?

A crossfade can hide amplitude seams, but it does not solve semantic drift. Prefix locking tries to make adjacent chunks agree before vocoding.

What this really is

A sampling-policy contribution on top of the diffusion model, not just a post-processing trick.

Lab 4: Diagnostics from the Full-Song Test



Current full-song test

duration: 160 s

chunks: 64

boundary mel MSE mean: 0.00183

boundary discontinuity mean: 2.87 dB

Interpretation

Overlap handling is helping. The remaining problem is mostly gradual warble and static accumulation rather than obvious seam cuts. Long-form quality depends on chunk policy, anchoring, vocoding, and stabilization together, not just on the base diffusion model.

Quantitative Summary Across Labs



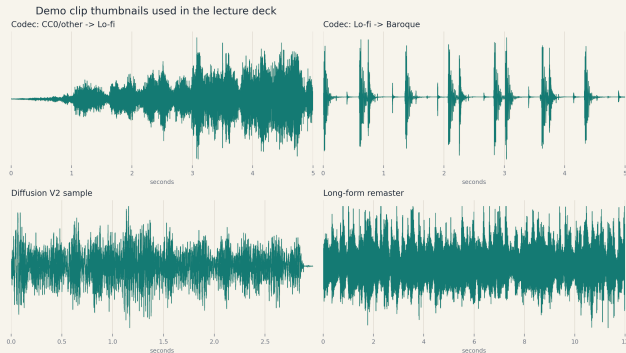
Already strong

Lab 1 and Lab 2 both clear their thresholds. The codec branch also clears the core short-form style and content gates.

Still open

Better diffusion audio quality, stronger long-form texture stability, and a finished blind listening study in Lab 5.

Audio Demo I: Short-Form Codec Examples



Clips from run1055

CC0 other to lo-fi

Audio clip 1 (5 s)

media/codec_src1_tgt3.mp3

hip-hop to open-domain other

Audio clip 2 (5 s)

media/codec_src2_tgt1.mp3

lo-fi to baroque

Audio clip 3 (5 s)

media/codec_src3_tgt0.mp3

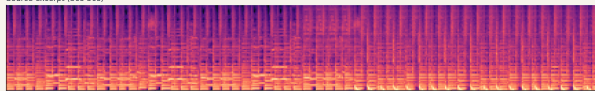
What to listen for

Does the melody survive? Is the target genre perceptible? Does the result sound like plausible audio rather than a synthetic artifact?

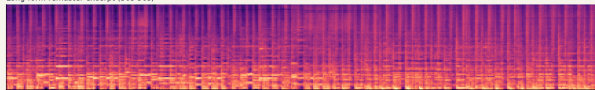
Audio Demo II: Diffusion and Long-Form Excerpts

Matched excerpt for long-form listening comparison

Source excerpt (30s-50s)



Long-form remaster excerpt (30s-50s)



Seconds in excerpt

Diffusion clip

Audio Diffusion V2 sample (3 s)

`media/diffusion_v2_epoch6_gen0.mp3`

Long-form A/B excerpt

Source excerpt, 30 s–50 s

Audio source (20 s)

`media/longform_source_excerpt.mp3`

Remastered excerpt, same segment

Audio remaster (20 s)

`media/longform_remaster_excerpt.mp3`

What to focus on

Diffusion clip: edit freedom and texture.

Long-form pair: continuity and artifact buildup.

Creative Elements and Novelty

Creative / novel components

1. Structure-first framing of genre remastering instead of shallow timbre swapping.
2. A staged pipeline with explicit deconstruction, calibration, reconstruction, and coherence stages.
3. Direct-output codec translation plus MERT-based conditioning.
4. Prefix-lock coherence constraints inside diffusion sampling.

Why these count

They are documented engineering responses to observed bottlenecks in the repo, not generic off-the-shelf invocations.

How to present this

Emphasize that the novelty is not one isolated block. It is the combination of representation learning, stronger conditional translation, and long-form sampling controls.

Creative-elements angle for CMPUT 414

The project contributes new engineering structure and tuning decisions rather than merely re-running a known model with default settings.

Limitations and Lab 5

Current limitations

1. Dataset-source leakage is still a genre-label risk.
2. Diffusion quality currently lags codec transfer on short-form reliability.
3. Long-form audio still accumulates warble and static under stronger edits.

Lab 5 next step

Run a blind listening study against simpler filter-style baselines and report both subjective genre authenticity and objective realism proxies.

What the final study should answer

Can listeners identify the target genre? Can they still recognize the source melody? Do they prefer our remaster to a simpler baseline?

Why this is the right ending

At this point the key open question is no longer whether the architecture runs. It is whether people actually hear the intended effect.

Conclusion

Lecture takeaway

Serious music style transfer is not one trick. It needs a representation story, a conditioning story, a synthesis story, and a coherence story.

Case-study takeaway

DGGR already clears its short-form technical gates and extends into measurable long-form coherence. The remaining gap is mainly perceptual refinement and formal human evaluation.

One-sentence summary

We started from the general problem of meaningful genre transfer and built a repo-backed system that implements one defensible answer end-to-end.

Discussion prompts

Codec editing, diffusion, or hybrid?
How should music evaluation balance taste with repeatability?

Backup: Exact Codec Branch Dimensions

Component	Implementation detail
Input latents	EnCodec quantized embeddings, approximately $[B, 128, T]$
Condition vector	$z_c \in \mathbb{R}^{128}$, target style embedding $z_s \in \mathbb{R}^{128}$, noise $\eta \in \mathbb{R}^{32}$
Hidden width	256 channels
Residual stack	10 FiLM-conditioned residual Conv1D blocks
Dilations	$[1, 2, 4, 8, 1, 2, 4, 8, 1, 2]$
Normalization	GroupNorm with 8 groups
Output mode	residual or direct-output translation; best run uses direct-output
Adversarial path	multi-scale waveform discriminator with hinge loss and feature matching

Backup: Exact Diffusion Branch Dimensions

Component	Implementation detail
Input tensor	$[B, 15, 80, 432] = 1 \text{ noisy mel} + 12 \text{ chroma} + 1 \text{ onset} + 1 \text{ beat channel}$
Output tensor	$[B, 1, 80, 432]$ velocity/noise prediction
UNet channels	$[64, 128, 256, 256]$
Residual blocks	2 per resolution level
Attention	lower resolutions in the current code path
Condition split	time and content via FiLM, style via dedicated AdaIN
Schedule	cosine beta schedule, DDIM sampling, EMA shadow model
Waveform path	BigVGAN-compatible mel extraction and waveform reconstruction

Backup: Repo Evidence Used for This Lecture

Core repo-backed evidence

Lab 1 explanations and audits in docs/explanation/lab1_deconstruction_encoder.md and saves/lab1_run_combo_af_gate_exit_v2/.

Lab 2 calibration artifacts in saves/lab2_calibration/lab2_20260211_015118_lda_cleanup_v2/.

Codec branch code in dggr/lab3_codec_models.py; best metrics in saves2/lab3_codec_transfer/run1055/.

Diffusion branch code in dggr/lab3_diffusion_model.py; training history in saves2/lab3_diffusion/run_d002/.

Long-form coherence code in lab_3/run_lab4_longform_coherence.py; diagnostics in saves2/lab4_longform_coherence/fullsong_test/.

Visual and media assets

Charts and spectrograms were generated locally from repo artifacts using presentation/generate_figures.py. Audio demo clips were converted from existing repo outputs into presentation/media/. OpenMoji icons are used through the local LaTeX package.

Backup: Lecture References

Dai et al. (2018), Aucouturier and Pachet (2003), Ghosal and Kolekar (2018), Hung et al. (2019), Yang et al. (2019), Roberts et al. (2018), Donahue et al. (2019), Gulrajani et al. (2017), Engel et al. (2019), Lee et al. (2023), Perez et al. (2018), Ho et al. (2020), Ho and Salimans (2021), Borsos et al. (2023), Meng et al. (2022).